

MODULE 21 · EXAM STRATEGY - ALL DOMAINS

Confusion Pairs

The service pairs that actually appear together as options, and the word that separates each.

SCOPE All domains TYPE Tiebreakers MODULE 21 of 26

These are not pairs that **sound** similar.
They are pairs that measurably **appear together** as options.

CloudTrail or CloudWatch?

AWS CLOUDTRAIL

Who did what

- Records API calls and account activity
- Answers which user changed a security group
- Auditing, governance, compliance
- The subject is an action by an identity

AMAZON CLOUDWATCH

How it is behaving

- Collects metrics, logs and alarms
- Answers whether the instance failed a status check
- Monitoring and operational health
- The subject is a measurement over time

THE TIEBREAKER

Who did it means CloudTrail. How is it performing means CloudWatch. If the question names a user, it is CloudTrail.

AWS Shield or AWS WAF?

AWS SHIELD

Volume, not content

- DDoS protection at layers 3 and 4
- Standard is free and automatic for everyone
- Advanced adds response team and cost protection
- It does not read your requests

AWS WAF

Content, not volume

- Filters HTTP and HTTPS requests at layer 7
- Rules on SQL injection, cross-site scripting, IP, geography
- Attaches to CloudFront, ALB, API Gateway
- It reads and judges each request

THE TIEBREAKER

Flooding means Shield. Inspecting the request means WAF. Both can protect the same Application Load Balancer.

GuardDuty or Inspector?

AMAZON GUARDDUTY

Someone is doing something

- Continuous threat detection from logs and network flow
- Unusual API calls, crypto-mining, compromised credentials
- Finds active malicious behaviour
- Reads what is happening now

AMAZON INSPECTOR

Something is exposed

- Scans workloads for software vulnerabilities
- Unintended network exposure of EC2, ECR images, Lambda
- Finds weaknesses before anyone uses them
- Reads what could be exploited

THE TIEBREAKER

Behaviour and threats means GuardDuty. Vulnerabilities and patches means Inspector. Active attack versus latent weakness.

Macie or GuardDuty?

AMAZON MACIE

What the data is

- Machine learning and pattern matching over Amazon S3
- Discovers personally identifiable information
- Classifies and flags sensitive data at rest
- Cares about content, not behaviour

AMAZON GUARDDUTY

What is being done

- Threat detection across account activity
- Unusual API calls and network behaviour
- Flags compromise, not classification
- Cares about behaviour, not content

THE TIEBREAKER

If the question says sensitive data, PII, or S3 content, it is Macie. Macie is the only one that looks inside your data.

AWS Config or AWS CloudTrail?

AWS CONFIG

What it looks like now, and before

- Records resource configuration over time
- Evaluates rules for compliance
- Answers was this bucket ever public
- The subject is a configuration state

AWS CLOUDTRAIL

Who changed it, and when

- Records API calls and identities
- Answers who made the bucket public
- Auditing of actions
- The subject is an event

THE TIEBREAKER

Config answers what the configuration was. CloudTrail answers who changed it. Compliance rules always mean Config.

Trusted Advisor or Config?

AWS TRUSTED ADVISOR

AWS's opinion of your account

- Checks AWS best practice automatically
- Cost, performance, security, fault tolerance, service limits
- You did not write the checks
- Recommendations you may ignore

AWS CONFIG

Your rules about your account

- Rules you enable or author
- Evaluates resource configuration against them
- You decide what compliant means
- Findings against your own standard

THE TIEBREAKER

Best practice recommendations means Trusted Advisor. Compliance with our policy means Config.

Security group or network ACL?

SECURITY GROUP

Instance level, stateful

- Attaches to an instance's network interface
- Return traffic allowed automatically
- Allow rules only, no deny rules
- All rules evaluated together

NETWORK ACL

Subnet level, stateless

- Attaches to a subnet
- Return traffic needs an explicit rule
- Allow and deny rules both available
- Rules evaluated in numbered order

THE TIEBREAKER

Instance means security group. Subnet means network ACL. Need to block a specific address means network ACL, because security groups cannot deny.

AWS Direct Connect or Amazon Connect?

AWS DIRECT CONNECT

A private network link

- Dedicated physical connection to AWS
- Consistent bandwidth, bypasses the internet
- Hybrid networking
- Nothing to do with customers

AMAZON CONNECT

A cloud contact centre

- Voice and chat customer service
- Pay as you go, agents can be remote
- Build contact flows without infrastructure
- Nothing to do with networking

THE TIEBREAKER

These share a word and nothing else. Direct Connect is a cable. Amazon Connect is a call centre.

IAM or IAM Identity Center?

AWS IAM

Identity inside one account

- Users, groups, roles and policies
- Permissions within a single AWS account
- Long-lived users, or roles for temporary access
- The default answer for permissions questions

AWS IAM IDENTITY CENTER

Single sign-on across many

- Central access to multiple AWS accounts
- Connects an existing identity provider
- Workforce single sign-on
- Formerly AWS Single Sign-On

THE TIEBREAKER

One account means IAM. Multiple accounts, workforce sign-on, or an external directory means IAM Identity Center.

AWS Budgets or Cost Explorer?

AWS BUDGETS

Looks forward

- Set a threshold and get alerted
- Cost, usage, Reserved Instance and Savings Plans budgets
- Acts before you overspend
- The subject is a limit

AWS COST EXPLORER

Looks backward

- Visualise and analyse historical cost and usage
- Filter and group by service, tag or linked account
- Explains what already happened
- The subject is a trend

THE TIEBREAKER

Alert me when we exceed means Budgets. Show me what we spent means Cost Explorer.

Ten more pairs, one line each

THE PAIR	THE SEPARATOR
RDS or DynamoDB	Relational and SQL means RDS. Key-value, NoSQL, single-digit millisecond means DynamoDB.
DynamoDB or Redshift	Operational lookups means DynamoDB. Analytics over a warehouse means Redshift.
Aurora or RDS	Aurora is an RDS engine, MySQL and PostgreSQL compatible, higher performance. Oracle or SQL Server means RDS.
EFS or FSx	Linux and NFS means EFS. Windows and SMB means FSx for Windows File Server.
EBS or S3	A disk attached to one instance means EBS. Objects over HTTP at any scale means S3.
ECS or EKS	AWS's own orchestrator means ECS. Kubernetes specifically means EKS.
Fargate or EC2 launch type	No servers to manage means Fargate. Control of the instances means EC2.
Region or Availability Zone	Geographic area means Region. Isolated data centres inside one means Availability Zone.
Availability Zone or edge location	High availability inside a Region means AZ. Low latency content delivery means edge location.
CloudFormation or Elastic Beanstalk	Define any infrastructure as code means CloudFormation. Upload an app and let AWS run it means Beanstalk.

ON THE EXAM

Read this table out loud once a day in the week before your exam. It is the highest-density revision in the course.

A security team is investigating an incident. Overnight, an Amazon S3 bucket that had been private became publicly readable. By morning it was private again.

The team needs to establish two things: which IAM identity made the change, and whether the bucket has ever been publicly readable at any point in the last year.

Which two services should the team use? (Choose two.)

- ☐ A AWS CloudTrail
- ☐ B Amazon CloudWatch
- ☐ C AWS Config
- ☐ D Amazon Macie
- ☐ E AWS Trusted Advisor

◆ PAUSE AND DECIDE

Which two services should the team use? (Choose two.)

- A

AWS CloudTrail
Records the API call and the identity that made it. Answers the who.
- B

Amazon CloudWatch
Metrics and logs about performance. It does not record who changed a bucket policy.
- C

AWS Config
Keeps configuration history for the resource. Answers has it ever been public.
- D

Amazon Macie
Would tell them whether the bucket holds sensitive data. Useful next, but not asked here.
- E

AWS Trusted Advisor
Flags public buckets as a current finding, but keeps no history and names no identity.

THE SIGNAL

Two questions, two services. Who did it is CloudTrail. What state was it in over time is Config.

A public web application behind an Application Load Balancer is being hit from thousands of addresses. Analysis shows two separate problems: a flood of meaningless traffic saturating the connection, and a smaller set of requests attempting SQL injection against the login form.

The company wants to address both, and wants to know what it already has.

Which combination addresses both problems?

- ☐ A AWS WAF for both problems
- ☐ B AWS Shield for the flood; AWS WAF for the injection attempts
- ☐ C Security groups for the flood; AWS Shield for the injection attempts
- ☐ D Amazon GuardDuty for both problems

◆ PAUSE AND DECIDE

Which combination addresses both problems?

- A

 AWS WAF for both problems
WAF inspects request content but does not absorb volumetric attacks at layers 3 and 4.
- B

 AWS Shield for the flood; AWS WAF for the injection attempts
Volume to Shield, content to WAF. Shield Standard is already active at no charge.
- C

 Security groups for the flood; AWS Shield for the injection attempts
Both halves wrong. Security groups cannot deny, and Shield never reads request content.
- D

 Amazon GuardDuty for both problems
GuardDuty detects and reports threats. It is a detection service, not a protection service.

THE SIGNAL

Two problems at two layers. Volume is layer 3 and 4, so Shield. Request content is layer 7, so WAF.

A company runs a fleet of Windows servers on premises that share files using the SMB protocol. It is migrating to AWS and wants a fully managed file system that the existing Windows applications can mount without modification.

A second team runs Linux application servers that need a shared file system mounted by many instances at once.

Which two services should the company use? (Choose two.)

- ☐ A Amazon S3
- ☐ B Amazon EBS
- ☐ C Amazon EFS
- ☐ D Amazon FSx for Windows File Server
- ☐ E AWS Storage Gateway

◆ PAUSE AND DECIDE

Which two services should the company use? (Choose two.)

- A

 Amazon S3
Object storage over HTTP. It is not a mountable file system, so applications would need rewriting.
- B

 Amazon EBS
Block storage attached to a single instance. It cannot be shared across many instances.
- C

 Amazon EFS
Managed NFS file system, mountable by many Linux instances at once.
- D

 Amazon FSx for Windows File Server
Managed file system with native SMB support. Windows applications mount it unmodified.
- E

 AWS Storage Gateway
Bridges on-premises to AWS storage. The scenario is migrating, not staying hybrid.

THE SIGNAL

SMB and Windows means FSx. NFS and Linux means EFS. Without modification rules out object storage entirely.

A finance director wants a monthly report of AWS spend broken down by department. Each department has its own AWS account, and all accounts sit under one organisation with consolidated billing.

Separately, the director wants an email whenever any single department's forecast monthly spend exceeds its allocation.

Which two services meet these requirements? (Choose two.)

- ☐ A AWS Cost Explorer
- ☐ B AWS Budgets
- ☐ C AWS Cost and Usage Report
- ☐ D AWS Pricing Calculator
- ☐ E AWS Cost Anomaly Detection

◆ PAUSE AND DECIDE

Which two services meet these requirements? (Choose two.)

- A

AWS Cost Explorer
Analyses historical cost, grouped and filtered by linked account. Exactly the departmental breakdown.
- B

AWS Budgets
Sets a threshold per account and alerts on forecast or actual spend crossing it.
- C

AWS Cost and Usage Report
The most detailed data available, but it is a raw file for analysis, not a report or an alert.
- D

AWS Pricing Calculator
Estimates the cost of things you have not built yet. It never sees your actual spend.
- E

AWS Cost Anomaly Detection
Alerts on unusual spikes, not on crossing a planned allocation. Different trigger.

THE SIGNAL

Two requirements again. Backward-looking breakdown is Cost Explorer. Forward-looking threshold alert is Budgets.

A retailer needs two things from its data. Its website must look up a customer's basket by customer identifier, thousands of times a second, in single-digit milliseconds.

Its analysts separately need to aggregate four years of completed orders to produce quarterly revenue reports.

Which two services should the retailer use? (Choose two.)

- ☐ A Amazon RDS
- ☐ B Amazon DynamoDB
- ☐ C Amazon Redshift
- ☐ D Amazon ElastiCache
- ☐ E Amazon Neptune

◆ PAUSE AND DECIDE

Which two services should the retailer use? (Choose two.)

- A

 Amazon RDS
Relational and capable, but neither requirement needs joins, and it will not match DynamoDB at that lookup rate.
- B

 Amazon DynamoDB
Retrieval by a single key, at any scale, in single-digit milliseconds. The operational half exactly.
- C

 Amazon Redshift
Aggregating four years of rows for a report is columnar analytical work. The analytical half exactly.
- D

 Amazon ElastiCache
Would speed repeated identical reads, but baskets differ per customer, so there is little to cache.
- E

 Amazon Neptune
Graph relationships. Nothing here traverses connections between entities.

THE SIGNAL

Who is waiting decides it. The website waits for DynamoDB; an analyst waits for Redshift.

A platform team runs containerised services and maintains a large library of existing Kubernetes manifests it wants to keep using. It has decided it no longer wants to patch or size any EC2 instances.

It also needs somewhere to store its container images.

Which two services should the team use for orchestration and compute? (Choose two.)

- ☐ A Amazon ECS
- ☐ B Amazon EKS
- ☐ C AWS Fargate
- ☐ D Amazon ECR
- ☐ E The EC2 launch type

◆ PAUSE AND DECIDE

Which two services should the team use for orchestration and compute? (Choose two.)

- A

 Amazon ECS
A capable orchestrator, but it does not run Kubernetes manifests. The existing library would be wasted.
- B

 Amazon EKS
Managed Kubernetes, so existing manifests and tooling keep working.
- C

 AWS Fargate
Removes the instances entirely, which is the second requirement. It works with EKS as well as ECS.
- D

 Amazon ECR
Genuinely needed for the images, but the question asks about orchestration and compute. ECR does neither.
- E

 The EC2 launch type
Keeps the instances the team explicitly no longer wants to patch or size.

THE SIGNAL

Two separations in one question: ECS against EKS, and Fargate against EC2. Each clause decides one of them.

A company already runs its web application across three Availability Zones behind a load balancer. Availability has been excellent and nothing has failed in a year.

Users in Asia and South America still report that pages take several seconds to load, while users near the Region are unaffected.

What should the company do?

- ☐ A Add a fourth Availability Zone
- ☐ B Use Amazon CloudFront to cache content at edge locations
- ☐ C Move the application to a larger instance type
- ☐ D Enable Multi-AZ on the load balancer

◆ PAUSE AND DECIDE

What should the company do?

- A

 Add a fourth Availability Zone
More availability inside the same Region. Availability is not the problem and the zones are all equally distant.
- B

 Use Amazon CloudFront to cache content at edge locations
Edge locations sit near those users, so content is served locally instead of crossing the world.
- C

 Move the application to a larger instance type
Users near the Region are unaffected, so the application is not slow. The distance is.
- D

 Enable Multi-AZ on the load balancer
A load balancer already spans zones, and this addresses availability rather than distance.

THE SIGNAL

Users near the Region are unaffected is the whole question. That sentence makes it geography, not capacity.

An insurance company is opening a customer service centre. It needs a pay-as-you-go system for handling inbound customer calls and chats, which remote agents will use from home.

The same company separately needs a dedicated private network link between its head office and AWS, with consistent bandwidth for a claims system.

Which two services meet these requirements? (Choose two.)

- | | |
|---|---|
| ☐ A Amazon Connect | ☐ B AWS Direct Connect |
| ☐ C AWS Site-to-Site VPN | ☐ D Amazon Chime |
| ☐ E AWS Transit Gateway | |

◆ PAUSE AND DECIDE

Which two services meet these requirements? (Choose two.)

- A

 Amazon Connect
A pay-as-you-go cloud contact centre for voice and chat, usable by remote agents.
- B

 AWS Direct Connect
A dedicated physical link with consistent bandwidth between premises and AWS.
- C

 AWS Site-to-Site VPN
Encrypted over the internet, so bandwidth varies. The requirement says consistent and dedicated.
- D

 Amazon Chime
Meetings and communications for your own staff, not a customer contact centre.
- E

 AWS Transit Gateway
A hub connecting many VPCs. Only one office link is described.

THE SIGNAL

Both correct answers contain the word Connect and share nothing else. That is the entire point of this question.

How to use this module

- ◆ **Rewatch this module last**, in the two days before your exam. It is designed for recall, not learning.
- ◆ **Learn the separator sentence, not the service description.** The separator is what you use under time pressure.
- ◆ When two options both look right, ask which pair you are looking at, then apply that pair's separator.
- ◆ Detection tells you. Protection stops it. GuardDuty, Inspector and Macie all only tell you.
- ◆ A correct real-world action that answers an unasked question is the commonest distractor on this exam.

ON THE EXAM

If you can state the separator for all twenty pairs from memory, you are ready for the service-selection questions.